

Final Project Guidelines

Course: Natural Language Processing

1. Overview

For the final project, students must **select one topic covered in class** and apply it to a **Chinese Natural Language Processing (NLP)** task.

The goal is to demonstrate your understanding of NLP concepts by implementing a practical system, experiment, or analysis that focuses on **Chinese text**.

Students are strongly encouraged to use **real-world datasets** to increase practical value and deepen understanding of real application scenarios.

2. Project Requirements

2.1 Select a Topic (We have done in midterm)

Choose **one** topic from the course lectures, such as (examples):

- Tokenization & Part-of-Speech tagging
- Text representation (TF-IDF, embeddings, BERT embeddings)
- Text classification (rule-based, SVM, K-Means, spaCy, LLM-based methods)
- Information extraction (regex, NER, similarity)
- Topic modeling
- Transformer-based tasks
- Question answering / NLU
- Generative AI applications (chatbot, text generation, text-to-SQL)

You may also propose your own topic if it is clearly NLP-related and approved by the instructor.

2.2 Implement a Chinese NLP Task

Your implementation should:

- Use **Chinese text** as the primary material
- Include **hands-on processing steps** (e.g., tokenization, embeddings, classification, extraction, visualization...)
- Show your working pipeline, methodology, and challenges encountered

You may use any appropriate tools or libraries (Python, spaCy, HuggingFace, scikit-learn, etc.).

2.3 Demonstrate Practical Applications

Your final report or presentation must describe:

- **A real or potential application** of your work
- Why this task matters
- What type of organizations or users might benefit
- How your approach solves a practical problem

This is an essential requirement and part of your grade.

Students are encouraged to use:

- News articles
- Social media text
- Open-source corpora
- Domain-specific data (finance, healthcare, education, etc.)
- Or datasets you build on your own
- https://zh.wikiversity.org/zh-hant/Taiwan_Tongues/%E8%87%BA%E7%81%A3%E8%8F%AF%E8%AA%9E%E9%96%8B%E6%BA%90%E8%AA%9E%E6%96%99%E5%BA%AB

2.4 Deliverables

Your submission must include:

1. **Code implementation** (Jupyter Notebook, Python scripts, etc.)
2. **Presentation slides** (No max limitation, please make sure that you can describe your project properly.)
3. **Oral presentation** (8-10 minutes)
4. **Dataset description** (source, preprocessing, limitations)

請確保英語口頭簡報內容為8-10分鐘，其餘補充資料、Code程式碼等等，可用簡報的Appendix呈現。

Presentation Format and guideline

1. Define the research question.
2. Propose the NLP method that you're using.
3. Show the processing of experiments and test.
4. Demonstrate the results and the solutions for your question.
5. Discuss the results and summarize your project.

請完整陳述你的期末專案所要解決的課題、描述你所使用的NLP方法、執行流程。並且呈現你針對該課題所得到的結果或解決方案，最後討論與總結你的期末報告。

3. Evaluation Criteria

Component	Weight	Description
Technical Complexity	20%	The level of difficulty in your implementation and the appropriateness of methods used.
Practical Applicability	30%	How useful the project is in real-world scenarios; clarity of the proposed application; relevance of the dataset.

Component	Weight	Description
Implementation Quality	30%	Code quality, correctness, completeness of the pipeline, experimental results, and clarity of analysis.
Oral Presentation	20%	Organization, clarity, demonstration quality, ability to explain the project and answer questions.

4. General Suggestions

- Choose a topic that genuinely interests you.
- Use real data whenever possible.
- Keep the scope reasonable—depth is more important than breadth.
- Clearly explain every step of your NLP pipeline.
- Show examples, visualizations, or demos during the presentation.

5. Academic Integrity

All work must be your own.

If you use external datasets, models, or code snippets, you must provide proper citation in the report.

If you have questions or wish to propose a custom topic, please contact the instructor in advance.